

Opgave 2

I et område $r \leq R$ giver en kuglesymmetrisk ladningsfordeling ophav til et elektrisk felt, $\mathbf{E}(r) = E_0 (2(r/R)^2 - r/R) \hat{r}$, hvor E_0 er en konstant.

a) Bestem ladningsfordelingens totalladning, Q .

$$(2.13) \oint_S \vec{E} \cdot d\vec{a} = \frac{Q_{enc}}{\epsilon_0}, \quad d\vec{a} = r^2 \sin\theta d\theta d\phi, \quad \vec{E}(R) = E_0 \left(2 \frac{R^2}{R^2} - \frac{R}{R} \right) \hat{r} = E_0$$

$$Q_{enc} = \epsilon_0 \int_0^{2\pi} \int_0^\pi E_0 R^2 \sin\theta d\theta d\phi = \underline{4\pi R^2 \epsilon_0 E_0}$$

b) Bestem et udtryk for ladningstætheden, $\rho(r)$, som funktion af r .

$$(2.14) \quad \vec{\nabla} \cdot \vec{E} = \frac{\rho}{\epsilon_0} \Leftrightarrow \rho(r) = \epsilon_0 \frac{1}{r^2} \frac{\partial}{\partial r} \left[r^2 E_0 \left(2 \frac{r^2}{R^2} - \frac{r}{R} \right) \right] = \epsilon_0 E_0 \left(\frac{8r}{R^2} - \frac{3}{R} \right)$$

Opgave 3

I et kubisk område af rummet, $0 < x < a$, $0 < y < a$, og $0 < z < a$, har det elektriske felt formen

$$\mathbf{E}(x, y, z) = \frac{E_0}{a^2} [(x^2 + z^2) \hat{x} + (y^2 + x^2) \hat{y} + (z^2 + y^2) \hat{z}],$$

hvor a og E_0 er konstanter.

a) Bestem ladningstætheden, $\rho(x, y, z)$, i området. Find dernæst den totale ladning, Q , indeholdt i området.

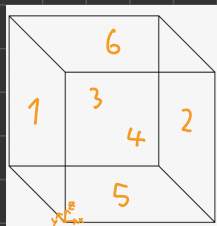
$$(2.14) \quad \vec{\nabla} \cdot \vec{E} = \frac{\rho}{\epsilon_0} \Leftrightarrow \rho(r) = \epsilon_0 \frac{E_0}{a^2} (2x + 2y + 2z) = \underline{\frac{2\epsilon_0 E_0}{a^2} (x + y + z)}$$

Gauss Sætning

$$(2.13) \quad \oint_S \vec{E} \cdot d\vec{a} = \frac{Q_{enc}}{\epsilon_0}, \quad Q = \epsilon_0 \oint_S \vec{E} \cdot \hat{n} d\vec{a} \stackrel{\downarrow}{=} \epsilon_0 \int_V (\vec{\nabla} \cdot \vec{E}) d\tau =$$

$$\epsilon_0 \int_0^a \int_0^a \int_0^a \frac{2E_0}{a^2} (x + y + z) dx dy dz = \frac{2\epsilon_0 E_0}{a^2} \cdot \frac{a^2}{2} a = \underline{3\epsilon_0 E_0 a^2}$$

Alternativt kan man regne $\oint_S \vec{E} \cdot d\vec{a}$ for hver af de 6 overflader og lægge sammen. For flade 4 gælder $y=0$, $d\vec{a} = -dx dz \hat{y}$



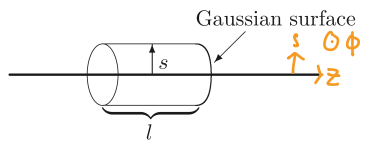
$$\oint_S \vec{E} \cdot \hat{n} d\vec{a} = -\frac{E_0}{a^2} \int_0^a \int_0^a (y^2 + x^2) dx dz = -\frac{E_0}{a^2} \int_0^a \int_0^a x^2 dx dz = -\frac{E_0}{a^2} \frac{a^3}{3} a = -\frac{E_0}{3} a^2$$

For flade 3 gælder $y=a$, $d\vec{a} = dx dz \hat{y}$

$$\oint_S \vec{E} \cdot \hat{n} d\vec{a} = \frac{E_0}{a^2} \int_0^a \int_0^a (y^2 + x^2) dx dz = \frac{E_0}{a^2} \int_0^a \int_0^a (a^2 + x^2) dx dz = \frac{E_0}{a^2} \left(a^3 + \frac{a^3}{3} \right) a = \frac{4}{3} E_0 a^2$$

Der er 3 par af flader så samlet giver $\oint_S \vec{E} \cdot d\vec{a} = 3 \left(\frac{4}{3} E_0 a^2 - \frac{E_0}{3} a^2 \right) = 3 E_0 a^2 \Rightarrow Q_{enc} = 3 \epsilon_0 E_0 a^2$

Problem 2.13 Find the electric field a distance s from an infinitely long straight wire, which carries a uniform line charge λ . Compare Eq. 2.9.



Cylindriske koordinater. Da ledningen er uendelig lang er den uændret under spejling vinkelret på z -aksen, så $E_z = 0$.

Ligeledes er den uændret under spejling parallelt med z -aksen, så $E_\phi = 0$.

(2.13) $\oint_S \vec{E} \cdot d\vec{a} = \frac{Q_{enc}}{\epsilon_0}$, $\vec{E} = E \hat{s}$, $da = s d\phi dz$, $Q_{enc} = \lambda l$

$$\oint_S \vec{E} \cdot d\vec{a} = E \int_0^{2\pi} \int_0^l s dz d\phi = E 2\pi l s = \frac{\lambda l}{\epsilon_0} \Leftrightarrow \underline{\vec{E} = \frac{\lambda}{2\pi s \epsilon_0} \hat{s}}$$

Magen til (2.9) med $z = s$

Problem 1.28 Calculate the line integral of the function $\vec{v} = x^2 \hat{x} + 2yz \hat{y} + y^2 \hat{z}$ from the origin to the point (1,1,1) by three different routes:

(a) $(0, 0, 0) \rightarrow (1, 0, 0) \rightarrow (1, 1, 0) \rightarrow (1, 1, 1)$;

(b) $(0, 0, 0) \rightarrow (0, 0, 1) \rightarrow (0, 1, 1) \rightarrow (1, 1, 1)$;

(c) The direct straight line.

(d) What is the line integral around the closed loop that goes *out* along path (a) and *back* along path (b)?

a) Linjeintegrale i 3 dele: (i) $y = z = 0$, $d\vec{l} = dx \hat{x}$. $\vec{v} \cdot d\vec{l} = x^2 dx$

$$\int \vec{v} \cdot d\vec{l} = \int_0^1 x^2 dx = \frac{1}{3}. \text{ (ii) } x = 1, z = 0, d\vec{l} = dy \hat{y}. \vec{v} \cdot d\vec{l} = 2yz dy = 0. \text{ (iii) } x = y = 1, d\vec{l} = dz \hat{z}$$

$$\vec{v} \cdot d\vec{l} = y^2 dz = 1 dz. \int \vec{v} \cdot d\vec{l} = \int_0^1 dz = 1. \oint \vec{v} \cdot d\vec{l} = 1 + \frac{1}{3} = \underline{\frac{4}{3}}$$

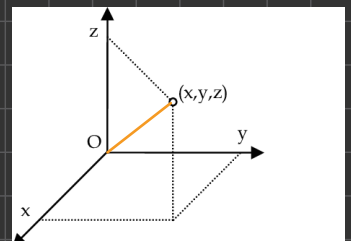
b) 3 dele igen (i) $x = y = 0$, $d\vec{l} = dz \hat{z}$. $\vec{v} \cdot d\vec{l} = y^2 dz = 0$. (ii) $x = 0, z = 1, d\vec{l} = dy \hat{y}$.

$$\vec{v} \cdot d\vec{l} = 2yz dy = 2y dy. \int \vec{v} \cdot d\vec{l} = \int_0^1 2y dy = 1. \text{ (iii) } y = z = 1, d\vec{l} = dx \hat{x}. \vec{v} \cdot d\vec{l} = x^2 dx.$$

$$\int \vec{v} \cdot d\vec{l} = \int_0^1 x^2 dx = \frac{1}{3}. \oint \vec{v} \cdot d\vec{l} = 1 + \frac{1}{3} = \underline{\frac{4}{3}}$$

c) $x = y = z$, $dx = dy = dz$, $d\vec{l} = dx \hat{x} + dy \hat{y} + dz \hat{z}$. $\vec{v} \cdot d\vec{l} = x^2 dx +$

$$2yz dy + y^2 dz = x^2 dx + 2x^2 dx + x^2 dx = 4x^2 dx. \int \vec{v} \cdot d\vec{l} = \int_0^1 4x^2 dx = \underline{\frac{4}{3}}$$



d) $\oint \vec{v} \cdot d\vec{l} = \frac{4}{3} - \frac{4}{3} = 0$

Problem 1.13 Let $\mathbf{r}$ be the separation vector from a fixed point (x', y', z') to the point (x, y, z) , and let r be its length. Show that

(a) $\nabla(r^2) = 2\mathbf{r}$.

(b) $\nabla(1/r) = -\mathbf{r}/r^3$.

(c) What is the general formula for $\nabla(r^n)$?

a) $\mathbf{r} = (x-x')\hat{x} + (y-y')\hat{y} + (z-z')\hat{z}$. $r = \sqrt{(x-x')^2 + (y-y')^2 + (z-z')^2}$

$$\nabla(r^2) = \frac{\partial}{\partial x} [(x-x')^2 + (y-y')^2 + (z-z')^2] \hat{x} + \frac{\partial}{\partial y} [(x-x')^2 + (y-y')^2 + (z-z')^2] \hat{y} + \frac{\partial}{\partial z} [(x-x')^2 + (y-y')^2 + (z-z')^2] \hat{z} = 2(x-x')\hat{x} + 2(y-y')\hat{y} + 2(z-z')\hat{z} = 2\mathbf{r}$$

b) $\nabla\left(\frac{1}{r}\right) = \frac{\partial}{\partial x} \left[((x-x')^2 + (y-y')^2 + (z-z')^2)^{-1/2} \right] \hat{x} + \frac{\partial}{\partial y} \left[((x-x')^2 + (y-y')^2 + (z-z')^2)^{-1/2} \right] \hat{y} + \frac{\partial}{\partial z} \left[((x-x')^2 + (y-y')^2 + (z-z')^2)^{-1/2} \right] \hat{z}$

$$= \frac{-1}{2} r^{-3} 2(x-x')\hat{x} + \frac{-1}{2} r^{-3} 2(y-y')\hat{y} + \frac{-1}{2} r^{-3} 2(z-z')\hat{z} = -r^{-3} ((x-x')\hat{x} + (y-y')\hat{y} + (z-z')\hat{z}) = -r^{-3} \mathbf{r} = -r^{-3} r \hat{r} = \underline{\underline{-\frac{\hat{r}}{r^2}}}$$

c) $\frac{\partial}{\partial x} (r^n) = n r^{n-1} \left(\frac{\partial}{\partial x} r \right) = n r^{n-1} \frac{1}{2r} 2(x-x') = n r^{n-2} r_x$

$$\nabla(r^n) = n r^{n-2} (r_x \hat{x} + r_y \hat{y} + r_z \hat{z}) = n r^{n-2} \mathbf{r} = \underline{\underline{n r^{n-1} \hat{r}}}$$

Problem 1.44. Evaluate the following integrals:

(a) $\int_2^6 (3x^2 - 2x - 1) \delta(x-3) dx$.

(b) $\int_0^5 \cos x \delta(x-\pi) dx$.

(c) $\int_0^3 x^3 \delta(x+1) dx$.

(d) $\int_{-\infty}^{\infty} \ln(x+3) \delta(x+2) dx$.

a) $\int_2^6 (3x^2 - 2x - 1) \delta(x-3) dx = 3 \cdot 3^2 - 2 \cdot 3 - 1 = 27 - 6 - 1 = \underline{\underline{20}}$

b) $\int_0^5 \cos x \delta(x-\pi) dx = \cos \pi = \underline{\underline{-1}}$

c) $\int_0^3 x^3 \delta(x+1) dx = \underline{\underline{0}}$ $x = -1$ er ikke på intervallet $[0, 3]$

d)

$$\int_{-\infty}^{\infty} \ln(x+3) \delta(x+2) dx = \ln(-2+3) = \ln 1 = \underline{0}$$

Problem 1.46

(a) Write an expression for the electric charge density $\rho(\mathbf{r})$ of a point charge q at $\mathbf{r}'$. Make sure that the volume integral of ρ equals q .

(b) What is the charge density of an electric dipole, consisting of a point charge $-q$ at the origin and a point charge $+q$ at $\mathbf{a}$?

(c) What is the charge density of a uniform, infinitesimally thin spherical shell of radius R and total charge Q , centered at the origin? [Beware: the integral over all space must equal Q .]

a)

$$\underline{\rho(\vec{r}) = q \delta^3(\vec{r} - \vec{r}')} \quad \text{Tjekker: } \int \rho d\tau = q \int \delta^3(\vec{r} - \vec{r}') d\tau = q \quad \checkmark$$

b)

$$\underline{\rho(\vec{r}) = -q \delta^3(\vec{r}) + q \delta^3(\vec{r} - \vec{a})}$$

This is exactly what the delta function $\delta^3(\mathbf{r} - \mathbf{r}')$ is built to do. By definition:

- $\delta^3(\mathbf{r} - \mathbf{r}') = 0$ for any $\mathbf{r} \neq \mathbf{r}'$ (zero density everywhere except at the point),

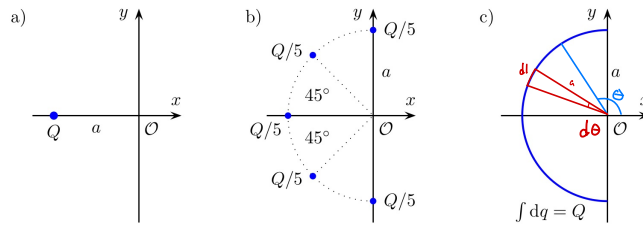
c)

$$\sigma = \frac{Q}{4\pi R^2} \quad \underline{\rho(\vec{r}) = \frac{Q}{4\pi R^2} \delta(r - R)}$$

$$\text{Tjekker: } \int \rho d\tau = \frac{Q}{4\pi R^2} \int_0^{2\pi} \int_0^\pi \int_0^\infty \delta(r - R) r^2 \sin\theta dr d\theta d\phi = \frac{Q}{4\pi R^2} 4\pi R^2 = Q \quad \checkmark$$

Opgave 1

Der undersøges tre forskellige tilfælde af ladningsfordelinger anbragt i xy -planet i et koordinatsystem. I alle tre tilfælde, som illustreret på figuren, befinder samtlige elementer af ladningsfordelingen sig i afstanden a fra koordinatsystemets begyndelsespunkt, $\mathcal{O}$.



a) En punktladning Q er anbragt på den negative x -akse i afstanden a fra $\mathcal{O}$. Bestem det elektriske felt i $\mathcal{O}$. Resultatet bedes givet på vektorform.

$$\vec{E} = \frac{1}{4\pi\epsilon_0} \frac{Q}{a^2} \hat{x}$$

b) Fem identiske punktladninger, $Q/5$, er anbragt på en cirkelbue med radius a centreret i $\mathcal{O}$ i punkter bestemt ved vinkler $\phi = -90^\circ, -45^\circ, 0^\circ, 45^\circ$ og 90° relativt til den negative x -akse. Bestem det elektriske felt i $\mathcal{O}$ på vektorform.

Af Symmetriårsager er vi kun interesseret i den del af feltet, der peger langs x -aksen, hvilket udelukker de to ladninger langs y -aksen.

$$\vec{E} = \frac{1}{4\pi\epsilon_0} \left(\frac{Q}{5} + 2 \cos(45^\circ) \frac{Q}{5} \right) \hat{x} = \frac{1}{4\pi\epsilon_0} \frac{Q}{5} \left(1 + 2 \frac{\sqrt{2}}{2} \right) \hat{x} = \frac{1}{4\pi\epsilon_0} \frac{Q(1+\sqrt{2})}{5} \hat{x}$$

c) Totalladningen Q er jævnt fordelt over en halvcirkel, som spænder anden og tredje kvadrant af xy -systemet. Halvcirklen, der er centreret i $\mathcal{O}$, har radius a . Bestem det elektriske felt i $\mathcal{O}$ på vektorform.

$$\vec{r} = \vec{0}, \quad \vec{r}' = a(\cos\theta\hat{x} + \sin\theta\hat{y}), \quad \vec{r} = \vec{r} - \vec{r}' = -\vec{r}', \quad r = a, \quad \hat{r} = \frac{-\vec{r}'}{a} = -\cos\theta\hat{x} - \sin\theta\hat{y}$$

$$\lambda = \frac{Q}{\pi a}, \quad dq = \lambda dl = \lambda a d\theta = \frac{Q}{\pi} d\theta, \quad \theta: \frac{\pi}{2} \rightarrow \frac{3\pi}{2}$$

$E_y = 0$ symmetri igen

$$(2.5) \quad \vec{E} = \frac{1}{4\pi\epsilon_0} \int \frac{1}{r^2} \hat{r} dq = \frac{1}{4\pi\epsilon_0} \int_{\frac{\pi}{2}}^{\frac{3\pi}{2}} \frac{1}{a^2} (-\cos\theta\hat{x} - \sin\theta\hat{y}) \frac{Q}{\pi} d\theta =$$

$$\frac{-1}{4\pi\epsilon_0} \int_{\frac{\pi}{2}}^{\frac{3\pi}{2}} \frac{Q}{\pi a^2} \cos\theta d\theta \hat{x} = \frac{-1}{4\pi\epsilon_0} \frac{Q}{\pi a^2} [\sin\theta]_{\pi/2}^{3\pi/2} \hat{x} = \frac{-1}{4\pi\epsilon_0} \frac{Q}{\pi a^2} (-1-1) \hat{x} = \frac{1}{4\pi\epsilon_0} \frac{2Q}{\pi a^2} \hat{x}$$

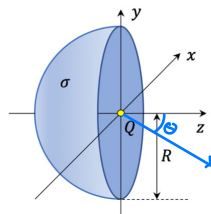
d) Bestem i de tre tilfælde, a), b), og c), det elektriske potential i $\mathcal{O}$. Nulpunktet for potentialet sættes i det uendelige.

$$a) (2.27) \quad V = \frac{1}{4\pi\epsilon_0} \frac{Q}{a} \quad b) \quad \frac{1}{4\pi\epsilon_0} 5 \frac{Q}{5a} = \frac{1}{4\pi\epsilon_0} \frac{Q}{a}$$

$$c) (2.28) \quad V = \frac{1}{4\pi\epsilon_0} \int \frac{1}{a} dq = \frac{1}{4\pi\epsilon_0} \int_{\frac{\pi}{2}}^{\frac{3\pi}{2}} \frac{Q}{\pi a} d\theta = \frac{1}{4\pi\epsilon_0} \frac{Q\pi}{\pi a} = \frac{1}{4\pi\epsilon_0} \frac{Q}{a}$$

Opgave 1

En homogen overfladeladningsfordeling med tæthed σ har form som den ene halvdel af en kugleskal med radius R , altså, som vist på figuren, $x^2 + y^2 + z^2 = R^2$, $z < 0$. En punktladning, Q , er placeret i origo, $(0, 0, 0)$.



Følgende spørgsmål bedes besvaret både symbolsk og numerisk, idet det opgives at $\sigma = 46,0 \text{ nC/cm}^2$, $Q = 0,75 \mu\text{C}$ og $R = 22,0 \text{ cm}$.

- a) Bestem Q 's elektrostatiske potentielle energi i feltet fra overfladeladningsfordelingen. Den potentielle energi sættes til nul i det uendeligt fjerne.

Alle ladningselementer dq befinder sig i samme afstand R fra Q .

Vi kan derfor kollapse dem til en samlet ladning $q = \int \sigma da = 2\pi R^2 \sigma$ med afstand R fra Q .

$$V = \frac{1}{4\pi\epsilon_0} \frac{q}{R} = \frac{1}{4\pi\epsilon_0} \frac{2\pi R^2 \sigma}{R} = \frac{R\sigma}{2\epsilon_0}$$

Alternativt bruges (2.30): $da = R^2 \sin\theta d\theta d\phi$, $\theta: \frac{\pi}{2} \rightarrow \pi$, $\phi: 0 \rightarrow 2\pi$

$$V = \frac{1}{4\pi\epsilon_0} \int \frac{\sigma}{R} da = \frac{1}{4\pi\epsilon_0} \int_0^{2\pi} \int_{\frac{\pi}{2}}^{\pi} \frac{\sigma}{R} R^2 \sin\theta d\theta d\phi = \frac{\sigma R}{4\pi\epsilon_0} 2\pi = \frac{R\sigma}{2\epsilon_0}$$

$$(2.39) \quad W = QV = \frac{QR\sigma}{2\epsilon_0} = \underline{4.3 \text{ J}}$$

- b) Anfør retningen af den elektriske kraft på Q . Bestem dernæst kraftens størrelse.

Af symmetriårsager må kraften være rettet langs z -aksen.

$\vec{r} = \vec{0}$, $\vec{r}' = R(\sin\theta \cos\phi \hat{x} + \sin\theta \sin\phi \hat{y} + \cos\theta \hat{z})$. Vi nøjes med z -delen

Så $\vec{r} = \vec{0} - r'_z \hat{z} = -R \cos\theta \hat{z}$, $r = R$, $\hat{r} = \cos\theta \hat{z}$, $da = R^2 \sin\theta d\theta d\phi$

$$(2.7) \quad \vec{E} = \frac{1}{4\pi\epsilon_0} \int_0^{2\pi} \int_{\frac{\pi}{2}}^{\pi} \frac{\sigma}{R^2} R^2 \sin\theta d\theta d\phi (-\cos\theta) \hat{z} = \frac{-\sigma}{4\pi\epsilon_0} 2\pi \left[-\frac{1}{2} \right] \hat{z} = \frac{\sigma}{4\epsilon_0} \hat{z}$$

$$\vec{F} = Q\vec{E} = \frac{Q\sigma}{4\epsilon_0} \hat{z} = \underline{9.75 \text{ N } \hat{z}}$$

Opgave 3

Der betragtes et område Ω i rummet, hvor ladningstætheden er nul. Begrund, hvilke af følgende udsagn er sande og hvilke falske:

- a) $\mathbf{E}_1(x, y, z) = A_1(y\hat{x} - x\hat{y} + d_1\hat{z})$ er et muligt elektrisk felt i Ω , idet A_1 og d_1 er reelle konstanter med enhederne henholdsvis V m^{-2} og m .

$$(2.14) \quad \bar{\nabla} \cdot \bar{\mathbf{E}}_1 = A_1(0 - 0 + 0) = 0 = \frac{\rho}{\epsilon_0} \Rightarrow \rho = 0$$

$$\bar{\nabla} \times \bar{\mathbf{E}}_1 = A_1(0\hat{x} + 0\hat{y} + (-1-1)\hat{z}) = -2A_1\hat{z}$$

$\bar{\mathbf{E}}_1$ er ikke et muligt felt, da $\bar{\nabla} \times \bar{\mathbf{E}}_1 \neq 0$ gælder for alle elektrostatiske felter.

- b) $V_2(r, \theta, \phi) = A_2(2\cos^2\theta - \sin^2\theta)r^2$ er et muligt elektrostatisk potential i Ω , idet A_2 er en reel konstant med enheden V m^{-2} .

$$(3.4) \quad \nabla^2 V_2 = A_2 \left[\frac{1}{r^2} \frac{\partial}{\partial r} (r^2 2r(2\cos^2\theta - \sin^2\theta)) + \frac{1}{r^2 \sin\theta} \frac{\partial}{\partial \theta} (r^2 \sin\theta (-4\cos\theta \sin\theta - 2\sin\theta \cos\theta)) \right] = A_2 [12\cos^2\theta - 6\sin^2\theta + \frac{-6}{\sin\theta} \frac{\partial}{\partial \theta} (\cos\theta \sin^2\theta)] = A_2 [12\cos^2\theta - 6\sin^2\theta - \frac{6}{\sin\theta} (2\cos^2\theta \sin\theta - \sin^3\theta)] = A_2 [12\cos^2\theta - 6\sin^2\theta - 12\cos^2\theta + 6\sin^2\theta] = 0$$

V_2 er et muligt elektrostatisk potential i Ω

Problem 2.21 Find the potential inside and outside a uniformly charged solid sphere whose radius is R and whose total charge is q . Use infinity as your reference point. Compute the gradient of V in each region, and check that it yields the correct field. Sketch $V(r)$.

$$(2.21) \quad V(\vec{r}) = - \int_0^{\vec{r}} \bar{\mathbf{E}} \cdot d\vec{l}, \text{ for } r \geq R: \bar{\mathbf{E}} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{r}, \text{ for } r < R: \bar{\mathbf{E}} = \frac{1}{4\pi\epsilon_0} \frac{q}{R^3} r \hat{r}$$

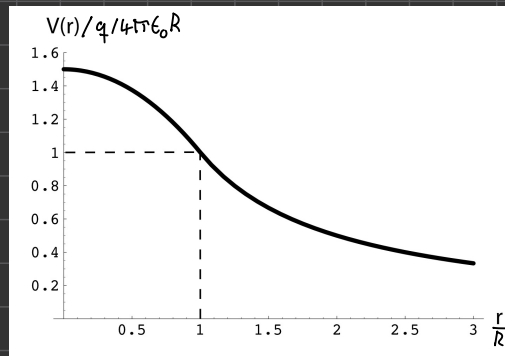
$$d\vec{l} = dr \hat{r}. \quad r \geq R: V = - \int_{\infty}^r \frac{1}{4\pi\epsilon_0} \frac{q}{r'^2} dr' = \frac{1}{4\pi\epsilon_0} \left[\frac{q}{r'} \right]_{\infty}^r = \frac{1}{4\pi\epsilon_0} \frac{q}{r}$$

$$r < R: V = - \int_{\infty}^R \frac{1}{4\pi\epsilon_0} \frac{q}{r'^2} dr' - \int_R^r \frac{1}{4\pi\epsilon_0} \frac{q}{R^3} r' dr' = \frac{1}{4\pi\epsilon_0} \left(\frac{q}{R} - \left[\frac{q r'^2}{2R^3} \right]_R^r \right) =$$

$$\frac{1}{4\pi\epsilon_0} \left(\frac{q}{R} - \left(\frac{q r^2}{2R^3} - \frac{q R^2}{2R^3} \right) \right) = \frac{1}{4\pi\epsilon_0} \frac{q}{2R} \left(3 - \frac{r^2}{R^2} \right)$$

$$r \geq R: \bar{\nabla} V = \frac{-1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{r} = -\bar{\mathbf{E}}$$

$$r < R: \vec{\nabla} V = \frac{-1}{4\pi\epsilon_0} \frac{q}{2R^2} \frac{2r}{R} \hat{r} = \frac{-1}{4\pi\epsilon_0} \frac{q}{R^3} r \hat{r} = -\vec{E}$$



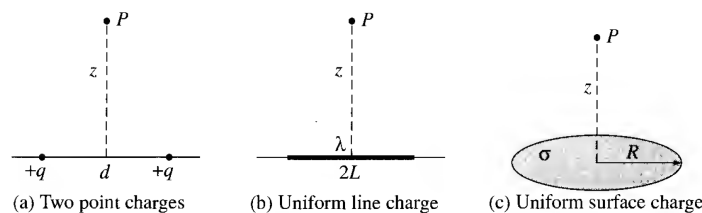
Problem 2.22 Find the potential a distance s from an infinitely long straight wire that carries a uniform line charge λ . Compute the gradient of your potential, and check that it yields the correct field.

$\vec{E} = \frac{1}{4\pi\epsilon_0} \frac{2\lambda}{s} \hat{s}$. Vi kan ikke sætte referencen til ∞ , da ledningen selv går til ∞ . Vi sætter den derfor til $s=a$. $d\vec{l} = ds \hat{s}$

$$V(s) = - \int_a^s \frac{1}{4\pi\epsilon_0} \frac{2\lambda}{s'} ds' = \frac{-1}{4\pi\epsilon_0} 2\lambda \left[\ln s' \right]_a^s = \frac{-1}{4\pi\epsilon_0} 2\lambda \ln\left(\frac{s}{a}\right) \quad \text{Her ses det at både } a=0 \text{ og } a=\infty \text{ ikke dner som referencepunkter.}$$

$$\vec{\nabla} V = \frac{-1}{4\pi\epsilon_0} 2\lambda \frac{\partial}{\partial s} \ln\left(\frac{s}{a}\right) \hat{s} = \frac{-1}{4\pi\epsilon_0} 2\lambda \frac{1}{s} \frac{1}{a} \hat{s} = \frac{-1}{4\pi\epsilon_0} \frac{2\lambda}{s} \hat{s} = -\vec{E} \quad \checkmark$$

Problem 2.25 Using Eqs. 2.27 and 2.30, find the potential at a distance z above the center of the charge distributions in Fig. 2.34. In each case, compute $\vec{E} = -\vec{\nabla} V$, and compare your answers with Prob. 2.2a, Ex. 2.1, and Prob. 2.6, respectively. Suppose that we changed the right-hand charge in Fig. 2.34a to $-q$; what then is the potential at P ? What field does that suggest? Compare your answer to Prob. 2.2b, and explain carefully any discrepancy.



a) (2.27) $r = \sqrt{\left(\frac{d}{2}\right)^2 + z^2}$ $V = \frac{1}{4\pi\epsilon_0} \frac{2q}{\sqrt{d^2/4 + z^2}}$

b) (2.30) $r = \sqrt{z^2 + x^2}$, $dl = dx$. $V = \frac{1}{4\pi\epsilon_0} \int_{-L}^L \frac{\lambda}{\sqrt{z^2 + x^2}} dx = \frac{\lambda}{4\pi\epsilon_0} \left[\ln(x + \sqrt{z^2 + x^2}) \right]_{-L}^L =$

$$\frac{\lambda}{4\pi\epsilon_0} \ln\left(\frac{L + \sqrt{z^2 + L^2}}{-L + \sqrt{z^2 + L^2}}\right) = \frac{\lambda}{4\pi\epsilon_0} \ln\left(\frac{L + \sqrt{z^2 + L^2}}{-L + \sqrt{z^2 + L^2}} \cdot \frac{L + \sqrt{z^2 + L^2}}{L + \sqrt{z^2 + L^2}}\right) = \frac{\lambda}{4\pi\epsilon_0} \ln\left(\frac{(L + \sqrt{z^2 + L^2})^2}{-L^2 + z^2 + L^2}\right) =$$

$$\frac{\lambda}{4\pi\epsilon_0} \ln\left(\frac{(L + \sqrt{z^2 + L^2})^2}{z^2}\right) = \frac{\lambda}{2\pi\epsilon_0} \ln\left(\frac{L + \sqrt{z^2 + L^2}}{z}\right)$$

$$c) (2.30) \quad r = \sqrt{r^2 + z^2}, \quad da = 2\pi r dr. \quad V = \frac{1}{4\pi\epsilon_0} \int_0^R \frac{\sigma}{\sqrt{r^2 + z^2}} 2\pi r dr = \frac{\sigma}{2\epsilon_0} \left[\sqrt{r^2 + z^2} \right]_0^R =$$

$$\frac{\sigma}{2\epsilon_0} (\sqrt{R^2 + z^2} - z)$$

$$a) \quad \vec{E} = -\vec{\nabla} V = \frac{-1}{4\pi\epsilon_0} 2q \left(\frac{-1}{z} \right) \frac{2z}{(z^2 + d^2/4)^{3/2}} \hat{z} = \frac{1}{4\pi\epsilon_0} \frac{2qz}{(z^2 + d^2/4)^{3/2}} \hat{z} \quad \checkmark$$

$$b) \quad \vec{E} = \frac{-\lambda}{2\pi\epsilon_0} \frac{\partial}{\partial z} \ln \left(\frac{L + \sqrt{z^2 + L^2}}{z} \right) \hat{z} = \frac{-\lambda}{2\pi\epsilon_0} \frac{-L}{z\sqrt{z^2 + L^2}} \hat{z} = \frac{1}{4\pi\epsilon_0} \frac{2\lambda L}{z\sqrt{z^2 + L^2}} \hat{z} \quad \checkmark$$

$$c) \quad \vec{E} = \frac{-\sigma}{2\epsilon_0} \left(\frac{z}{\sqrt{R^2 + z^2}} - 1 \right) \hat{z} = \frac{\sigma}{2\epsilon_0} \left(1 - \frac{z}{\sqrt{z^2 + R^2}} \right) \hat{z} \quad \checkmark$$

Hvis den højre ladning i a) sættes til $-q \Rightarrow V=0$. Man kunne fristes til så at sige $\vec{E} = -\vec{\nabla} V = \vec{0}$. Dette gælder dog kun langs z -aksen. Det gjorde vi også i a), men der var $E_x = E_y = 0$, dette er ikke længere tilfældet og det er derfor ikke tilstrækkeligt kun at kende potentialet langs z -aksen.